Weekly planning of hydropower in systems with large volumes and varying power generation: A literature review

Charlotta Ahlfors, Mikael Amelin

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***Abstract*—Hydropower is the world's largest source of renewable electricity generation and hydropower plants with reservoirs provide flexibility to the power systems. Efficient planning techniques improve the flexibility of the power systems and reduce carbon emissions, which is needed in power systems exposed to a rapid change. Renewable energy sources, such as wind and solar power, are increasing in the power systems. Hydropower plants have low operating costs and are used as base power. This paper reviews hydropower planning in different time frames and specifically focuses on weekly planning, i.e. hydropower planning for 1 to 3 weeks. Main conclusions in this study are that the term weekly planning is seldom used and the definitions in earlier studies of short term, mid term and long term planning, respectively, varies. The authors propose that weekly planning should belong to mid term planning.**

***Index Terms*—Hydropower, literature review, weekly planning, low carbon operation.**

I. Introduction

A. Background

Hydropower play a significant role to achieve the globally established UN sustainable goals and operating low carbon power systems [1]. The reason for this is that flexible generations, such as hydropower plants, can be used to keep the balance in the power system when use of renewable energy sources, e.g. photo voltaic generation and wind power, is rapidly increasing. In addition to increasing amount of renewable energy sources there is a growth in variable loads, electric vehicles and distributed generation, which adds more uncertainty to the power system. Hydropower plants with reservoirs can be used as flexible generation and compensate for power fluctuations in production and load for short time frames up to long time frames [2]. Hence, it is important to utilize the hydropower plants efficiently to cope with this development. Planning techniques based on optimization methods are developed, where the power producer aims to maximise the profit in the competitive decentralized electricity market. In literature, the planning is divided in plans for the coming years or seasons (long term planning), months (mid term planning) and plans for short time frame of the next few days (short term planning). This can be seen in figure 1. These terms are used as a well-known concept. Long term planning is used to plan the utilization of water in the largest reservoirs. These reservoirs are often located at the top of the river. Many parameters are stochastic, such as inflow to the reservoirs and electricity prices, since the planning is aimed for the coming years. The time resolution is usually rough since the output parameters are water values or a weekly discharge plan of power generation. The mid term planning is aimed for distribution of water in somewhat smaller reservoirs and therefore a more precise time resolution is used. The output parameters may also be water values. Short term planning is used to plan for the next few days and this planning is important to schedule the bidding to the spot market. Nowadays, variable renewable energy is implemented and stochastic behaviour should be considered in the hydropower models. In accordance with increased demand of integration of renewable energy in the power systems new planning techniques become interesting. A novelty to research is weekly planning and planning for this time frame is interesting to study since it cannot be found in literature and is interesting for development of low carbon operation power systems.

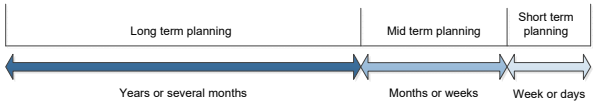


Fig. 1. Planning horizons within hydropower scheduling.

B. Motivation

The topic of hydropower planning is well investigated and various research questions have been studied under many years in different countries. Some of the works are summarized and discussed in literature reviews, which are presented in this section. First, some reviews are presented, which covers several aspects of hydro power planning. Literature reviews for long term, mid term and short term planning, respectively, are described.

In [3], a review of scheduling under uncertainty is presented. This study spans over many fields of science and the different techniques used to model uncertainties. Hydropower planning is one of the discussed topics. Some of these techniques are two-stage stochastic programming, parametric programming, fuzzy programming, chance constraint programming, robust optimization, conditional value-at-risk and other risk reducing techniques. In [4], a review of operation of hydropower plants is presented. It is concluded that the main parameters affecting the operation of a hydropower plant are head, discharge, turbine and the generators. In [5], a review on hydropower

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Charlotta Ahlfors is a PhD student at the Division of Electric Power and Energy Systems, Royal Institute of Technology (KTH), Stockholm, Sweden (e-mail: ahlfo@kth.se).

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plant models and control is performed. It summarizes the research work from 2005 and earlier of hydropower plant model development and its controller design.

In [6], from 2020, a comprehensive review is performed which spans over different time frames of hydropower scheduling. The different time spans for planning are defined as follows: short term 2-14 days, mid term 3-18 months and long term 1-5 years. An important conclusion and suggestion for future work is that scheduling of hydropower systems would be more valuable by considering other sustainable energy sources, such as wind and solar power, included in the optimization. Some models take this into account but it can be further developed.

A review of stochastic dual dynamic programming (SDDP) for long term hydropower planning is presented in [7]. It is concluded that the main strength of the method is to obtain reliable water values for all reservoirs and part of the computations can be done using parallel processing, due to the dual formulation of the optimization problem. Its weakness is the long computational times. In [8], a review is made of the application of the SDDP method applied to long term planning in the Nordic countries. For example, head variations and price modelling are discussed.

Several other methods for long term planning is reviewed in [9]. The methods are categorized in four main classes: Implicit stochastic optimization, explicit stochastic optimization, real-time control with forecasting and heuristic programming.

Recent hydropower optimization research and development using metaheuristic approaches are discussed in [10]. For example, physics based algorithms, gravitational search algorithms, and swarm based algorithms, such as particle swarm optimization, are reviewed and suitable for long term planning.

A review of short term planning in power systems with a large amount of wind power in the system is presented in [11]. The paper discusses deterministic model approaches as well as stochastic models. The conclusion in this paper is mainly that introducing several types of uncertainties lead to more realistic models but also models that are more difficult to study and analyse.

Earlier literature reviews have paid attention to the focus areas which have been mentioned. However, new research related to hydropower planning is of importance and weekly planning has not been in focus in earlier reviews. The scope of this paper is to summarize the reviews until today within hydropower planning, present selected representative studies within the different time frames of hydropower planning and present papers focusing in particular on weekly planning.

C. Contribution

This paper gives an overview of research that dates back to the 1960s until today within hydropower planning. Earlier literature reviews on hydropower planning are discussed. Studies within the different planning time frames; short term planning, mid term planning and long term planning, respectively, are discussed and summarized in tables. The tables summarize conclusions from the studies. Specifically, certain aspects of the models are highlighted, such as linear or non linear, stochastic behaviors and additional constraints, for example maintenance. Earlier studies within weekly planning of hydropower are presented. It is concluded that the definitions of the different planning time frames varies. A suggestion of definition is discussed, since it is important to communicate your research clearly and make it understandable to others. By categorizing weekly planning into mid term planning, it is possible to make adequate and tractable models. Another conclusion is that a few studies are performed within weekly planning of hydro power, and in particular systems with large amount of varying power generation. Some future research work in this area is proposed.

D. Structure of the paper

The paper is structured as follows: Section II presents selected representative papers of the different time frames of hydropower planning - long term planning, mid term planning and short term planning. Section III concludes the paper.

II. Hydropower planning

A. Long term planning

The aim of the long term planning is to make plans for one to several years in advance, where many parameters are uncertain, such as inflow and electricity prices. There are many examples of models where the stochastic behaviour is taken into account. The output from the long term planning is often water values or a weekly discharge plan of power generation. The output is used in mid term or short term planning directly.

In [12], water value methods that dates back to the 1960's are used. In this method aggregation of the reservoirs is done to make the computations efficient. These methods are based on stochastic dynamic programming (SDP). The SDP is compared to results from dynamic programming. A conclusion is that the hydropower plants can be operated more efficiently since less water is used and stored water can be used in the future. More water spillage is a consequence. The operational costs computed with dynamic programming where slightly higher than the expected operational costs computed with SDP. The conclusions are drawn based on a case study in Brazil. Another paper where the SDP method is used is [13] where the costs are reduced compared to when a standard Markovian model is used.

SDDP was introduced in [14], which allows for modeling of multi-reservoir systems without aggregation and leads to more realistic and detailed models. One major factor affecting the outcome of the SDDP method is a correct statistical model of the inflow and this is studied in [15] and [16]. Recent focus of the research on SDDP is on convergence properties, strategic bidding, risk aversion, emission reduction and integration of wind power and pumped-storage as outlined in [17].

Other examples of significant papers for long term planning are [18], [19] and and [20], which characteristics are described in Table I.

B. Mid term planning

Mid term planning models do not consider multi-year horizons but limit their attention to months. The models for mid term planning include more details and output such as water values.

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In [21], a model for the weekly scheduling of a hydropower system is developed, where bidding to both the day-ahead market and the reserve market are included. The model is constructed to protect against the risk of shortage of water and shortage of storage. It is shown that hydropower producers who participate in both markets have a significantly higher expected income than those only taking part in the day-ahead market. This method has the strength to account for both markets but at the same time it requires a proper scenario tree representation of the stochastic variables.

The model in [22] adds forward contracts as an ingredient in the operation of a hydropower plant, as well as pumped storage. It uses multistage stochastic quadratic programming. The results are water values and optimal forward bids to support the daily operation. A case study is performed, which shows that the forward trading contributes to a 20% increase in profit. Its main strength is that the model is both realistic and tractable. On the other hand, it assumes a linear dependence between the amount of commodities and market price and is also limited to a horizon of one year.

In [23], a mid term and long term electricity contract decomposition method is proposed that considers the cascade water quantity matching constraints. Mid term and long term power quantity contracts are decomposed by calculating the matching constraint of cascade water quantity. The method is tested in the Yunnan power spot market in China. The results show that the model is applicable for the Yunnan power spot market.

In [24], stochastic linear and non linear programming is used. The objective is to maximize the expected revenue of a producer, and the decision variables are generation and forward contracts in each period for each scenario. The parameters are limited by constraints on reservoir balance of water, upper and lower limits of power generation, bilateral contract and reservoir spillage.

In [25], a model is suggested of combined SDP and SDDP, where variables and functional relationships are linear. The main purpose is to fulfil the hydropower units operators' demands to get steady operation for the power grid. A case study is performed in Lysebotn in Norway. The study has resulted in a quantitative valuation including sales of capacity. However, the results from the paper may be case specific.

An extended version of the SDDP method is used in [26], stochastic dual dynamic integer programming (SDDiP). The method is applied to mid term planning. A test is performed on how to incorporate stagewise-dependent stochastic variables on the right-hand side of the objective function. The method has proven convergence in the nonconvex case. A case study is performed for a Norwegian hydropower producer. The case study shows that it is possible to solve the mid-term hydropower scheduling problem but time consuming to solve it to the optimal value.

In [27], the SDDiP method is applied in another Norwegian case study. The paper highlights that the method can be used to include nonconvex environmental constraints in mid term and long term planning. A case study is performed on a multireservoir system, with three reservoirs and corresponding hydropower plants. A particular nonconvex environmental constraint is modelled and a scheduling horizon of one year and weekly decision stages was applied.

C. Short term planning

The short term planning is performed for the nearest future and input parameters depend on the planning in the longer time horizons. The outcome of the planning is a production plan, usually with hourly resolution, of the amount of water that should be used in each power plant of the planning period and submitted to the day-ahead trading. Since the purpose is to find a detailed generation schedule, the models used for this planning include detailed information of the hydropower plants, turbines, and the topology in the river. Furthermore, the models include data for efficiency curves, discharge, inflow, spillage and reservoirs sizes.

In [28], a mixed integer stochastic model is developed that optimizes the bidding in a day-ahead market. The model treats the inflow and market price as stochastic. Through simulation, the results of the model are compared to results from a deterministic practice-based model, similar to what is used in the industry. The conclusion of the simulations is that the stochastic model performs better by 0.7%. This model has a number of positive features. For example, it is fast, can be used on a daily basis and is designed for the scheduling hierarchy used by producers today. The framework can be used to study other approaches to short term planning. The study is, however, limited to one system and needs further work to reach a reliable conclusion.

In [29], a stochastic short term planning model is proposed where day-ahead electricity prices and water inflows are modeled as scenarios based parameters on time series analysis. The model is constructed for a river with two cascaded hydropower plants. The schedule is determined for 7 days.

A non-linear dynamic programming model of a hydropower plant is presented in [30], where head and efficiency curves are treated as non-linear. The revenue of a producer that sells electricity in a pool-based electricity market is maximized. The plan is made for 1 day to 1 week with hourly resolution and the outcome is both the number of units in operation and the generation dispatch. A case study is made where the model is tested and an optimal weekly operation schedule has been obtained. The results of the model have been compared to results from a nonlinear programming model in 15 different daily cases. An increase in revenue is found with the dynamic programming model, ranging from 0.9% to 3.3%.

In [31], the non-linearity effects of head and efficiency curves are included in a stochastic model, where inflow and market price are considered uncertain. The non-linearity is handled using successive linear programming. The model is tested in three fictitious case studies and performs well. The paper investigates which factors influence the result the most but reaches no general conclusion.

Non-linearity is studied in [32]. A deterministic non-linear mixed integer optimization model is used to solve short term planning problems. The optimization problem is solved using Lagrangian relaxation leading to separable dual subproblems that are solved by the bundle method. A small hydropower system is used as a case study. The conclusion is that this

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approach is feasible but needs to be further studied on larger scale.

In [33], a mathematical model is proposed to simulate day-ahead markets of large-scale multi energy systems with high penetration of renewable energy towards 2050. Unit commitment (UC) is added to the model and the results are analysed. Including unit commitment constraints with integer variables lead to a more realistic behavior of the units at the cost of increasing the computational time. Relaxing integer variables significantly reduces the computational time without compromising the accuracy of the results. Hydro reservoirs were not modelled with UC in order to reduce the complexity of the problem. The novelty of the paper [33] involves developing a method for integrating renewable energy sources, planned maintenance scheduling, influence of unit commitment, multi-energy markets, full year hourly results, annual storage scheduling, several years of scenarios and a large scale system. The optimisations and simulations, except for variable renewable energy, are performed with the energy system model Balmorel (an open source, energy system tool, deterministic with bottom-up approach). The variable renewable energy hourly resolution time series data is simulated in CorRES and used as input to Balmorel. Data is aggregated into the regions that are used in Balmorel. A case study of the North Sea offshore Grid Denmark project was performed. Countries involved are Germany, Great Britain, Norway, Denmark, Belgium and Netherlands. The outcome of the tested scenario of electricity and heat show an increase in CO_2-price: 6.06, 76.70 and 130.38 €/ton in 2020, 2030 and 2050, respectively. The aggregated planned energy content along the year of hydro reservoirs with seasonal inflow of the Norwegian hydro storage reservoirs is investigated. Results from scenarios for 2020, 2030 and 2050 strengthen the importance of performing full year optimisation towards 2050. A conclusion from the study is that the case study is valid to other regions than the North Sea region, where high penetration of variable renewable energy is expected. The penetration of wind and solar is likely to challenge the need for balance in the system and the profitability of thermal units. In [34], the integration of wind power into a power system with limits on transmission capacity is studied. Stochastic linear programming is used and includes trading on the spot and regulating market. It is concluded from the study that coordination of hydropower production with wind power is beneficial for the utility of both power sources, great reduction in wind power curtailment and a more efficient usage of the transmission lines. Some model limitations are mentioned, such that the outcome of stochastic variables are assumed to be known in planning for the next day. Future work is suggested, for example, that a longer case study needs to be performed and the impact of coordination of start-ups for hydropower plants needs to be investigated.

In [35], a short term model, formulated as a mixed integer linear programming (MILP) problem, is presented. Power output and the number of units committed at each hydropower plant and hour of the day or week ahead are modeled. The aim of the planning is to maximise the hydropower efficiency while reducing start up or shut down costs and produce sufficient power to cover the load in the system. The method is to relax the hydraulic constraints, simulate the solution and reintroduce constraints that are violated in an iterative way. This leads to a computational efficient method that can be used in a large hydropower system.

In [36], a non-linear model for hydropower planning and scheduling of maintenance is proposed. The thermal generation complement is mimimized and the future value of stored water is maximized. The maintenance scheduling is modeled as a continuous variable. In [37], an algorithm is presented for the unit commitment problem that includes head and efficiency curves. The optimization problem is formulated as MILP problem. The optimization results in an accurate planning with reasonable computational time, since aggregation is used in the model.

A stochastic mixed integer model is developed in [38], that includes bidding to the day-ahead and balancing market and with start/stop decisions. Three implementations of the start/stop decisions (using none, real and binary variables) are tested. The conclusion is that there is only a small additional value in participating in the balancing market and there is a weak dependence of start/stop costs. The case study is performed on a small hydropower system in Norway and the power prices do not include the effect of a larger amount of intermittent power sources.

There are many other notable works on short term planning and a few are mentioned in this paragraph. The commitment problem in power systems with intermittent and fluctuating characteristics is studied in [39]. In [40], an overview is presented of formulations and optimization methods for unit-based short term hydro scheduling. A short term planning method is presented in [41] where Benders decomposition is used to solve a hydropower maintenance scheduling problem. Moreover, one notable short term planning paper is [42], which main characteristics and conclusions are summarized in Table III.

Table 1, summarizes earlier reviews on hydropower planning presented in Section I. Tables, 2 and 3 summarize the papers discussed in section II.

III. Conclusion

Power systems are rapidly changing into a large share of renewable power sources. This requires development of hydropower models, which are efficient and suitable for these power systems. The planning models used until today are denoted as long term planning, mid term planning and short term planning. From this study it can be concluded that the definitions of the different planning time frames are ambiguous and vary. A suggestion of definition is to use the terms; long term planning, mid term planning and short term planning as shown in Figure 1. This means that weekly planning of hydropower belongs to mid term planning.

Models for long term planning are performed in e.g. the Nordic countries, Brazil, New Zealand and China. Mid term planning models have been developed in e.g. Spain, Switzerland, China and Norway. Models for short term planning has been performed in e.g. the Nordic countries, Spain, Brazil and Canada. Since only a few models have been developed within

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the research area of weekly planning of hydropower, a suggestion for future work is to develop models which can be used by the Swedish power producers and which are useful in the electricity market. The models should consider large amount variable renewable power sources and stochastic behaviors such as inflows and electricity prices. Another suggestion is to combine the planning between different markets and integrate maintenance scheduling.

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TABLE I
THIS TABLE SUMMARIZES THE REVIEW PAPERS DISCUSSED IN SECTION I.

	Literature reviews	
Ref.	Title	Characteristics
[3]	Planning and Scheduling under Uncertainty: A review Across Multiple Sectors	Review that spans over many fields of science and the different techniques used to model uncertainties.
[4]	Operation of hydro power plants-a review	Review of operation of reservoir based as well as run of river hydropower plants.
[5]	Review on hydropower plant models and control	Review of hydropower plant models and control.
[6]	A Review of Optimization Algorithms in Solving Hydro Generation Scheduling Problems	Review of optimization algorithms for different time frames.
[7]	Handbook of Power Systems	Review of the SDDP method for long term hydropower planning.
[8]	Long-and medium-term operations planning and stochastic modelling in hydro-dominated power systems based on stochastic dual dynamic programming	Review of stochastic dual dynamic programming applied to hydropower scheduling in the Nordic countries.
[9]	Optimal Operation of Multireservoir Systems: State-of-the-Art-Review	Review of long term planning methods other than SDDP.
[10]	Optimization of the hydropower energy generation using Meta-Heuristic approaches: A review	Review of methods useful for long-term planning models.
[11]	The State-of-the-Art of the Short Term Hydro Power Planning with large Amount of Wind Power in the System	Review of short term planning works, deterministic and stochastic.

TABLE II
THIS TABLE SUMMARIZES THE PAPERS ON LONG- AND MID TERM PLANNING DISCUSSED IN SECTION II.

	Long term planning		
Ref.	Main characteristics	Case study location	Conclusions
[12]	Deterministic versus stochastic dynamic programming for long term hydropower scheduling.	Brazil	Deterministic and stochastic dynamic programming approaches in long term hydropower scheduling.
[13]	SDP, Markovian model for inflow aggregated over time scales.	Brazil	Reduced costs compared to standard Markovian model.
[14]	Optimal stochastic operations scheduling of large hydro-electric systems.	Brazil	SDDP method used for optimal operating strategies in a multi-reservoir hydroelectric system.
[15]	SDDP, large system, model for inflow.	Norway	SDDP is preferred to aggregate-disaggregate models.
[16]	SDDP, IFS model for inflow compatible with SDDP.	New Zealand	IFS model is an accurate representation of the inflow in two aspects.
[17]	Hybrid SDP/SDDP, inflow, energy price and reserve capacity stochastic.	Norway	Linearization error is pronounced in sales of energy and reserve capacity.
[18]	Formal optimization model compared to aggregated heuristic model.	Fictious	Both models give comparable results.
[19]	SDP, multi-stage problem is decomposed into two-stage optimization problems.	China	The method avoids the curse of dimensionality with good performance.
	Mid term planning		
Ref.	Main characteristics	Case study location	Conclusions
[20]	Flexible robust optimization dispatch for hybrid wind/photovoltaic/hydro/thermal power system.	China	Flexible robust optimization with adjustable uncertainty budget dispatch model built for a hybrid power system.
[21]	Stochastic MILP, day-ahead and reserve markets considered.	Spain	Higher expected income when participating in both day-ahead and reserve markets than just day-ahead.
[22]	Stochastic multi-stage quadratic model, pumped storage and forwards used	Switzerland	Trading with forwards increase profit.
[23]	Mid term and long term power quantity contracts are decomposed.	China	Tested model works well for the Yunnan power spot market.
[24]	Medium term power planning with bilateral contracts.	Norway	Stochastic linear and non-linear programming is used.
[25]	A case study on medium-term hydropower scheduling with sales of capacity.	Norway	SDP and SDDP are combined.
[26]	Nonconvex Medium-Term Hydropower Scheduling by Stochastic Dual Dynamic Integer Programming	Norway	The method works to solve to optimal value but is time consuming.
[27]	Nonconvex Environmental Constraints in Hydropower Scheduling	Norway	A particular nonconvex environmental constraint was modelled.

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TABLE III
THIS TABLE SUMMARIZES THE PAPERS ON SHORT TERM PLANNING DISCUSSED IN SECTION II.

	Short term planning		
Ref.	Main characteristics	Case study location	Conclusions
[28]	Mixed integer stochastic model, inflow and stochastic market price.	Norway	The model performs better than a deterministic practice-based industrial model.
[29]	Stochastic MILP, day-ahead price and inflow stochastic.	Norway	Profit and risk negatively correlated.
[30]	NLDP, head and efficiency curves non-linear.	Spain	Model considers many non-linearites and gives useful results.
[31]	Stochastic non-linear model solved with successive LP.	Fictious	The solution is close to optimal. Stochastic model performs better than deterministic.
[32]	Deterministic non-linear mixed integer optimization model solved with Lagrangian relaxation.	Brazil	The approach is feasible but needs to be further studied.
[33]	Mathematical model to simulate Day-ahead markets of large-scale multi-energy systems with large amount of renewable energy towards 2050.	North Sea region	The high penetration of wind and solar is likely to challenge the need for balancing in the system. Flexibility will be very important towards 2050.
[34]	Stochastic LP model with integrated wind power and trading.	Sweden	Coordinating hydro and wind power is beneficial.
[35]	MILP model with power output and unit commitment as variables.	Brazil	The model is efficient and generates a smooth generation schedule.
[36]	Non-linear model for power output and unit commitment.	Brazil	The model is compact and efficient but not convex.
[37]	MILP, unit commitment model.	Brazil	An accurate planning result within reasonable computational work.
[38]	Stochastic MIP commitment model with bids to day-ahead and balancing market.	Norway	Small additional revenue to participate in balancing market and weak dependence on start/stop costs.
[39]	MILP commitment model with large amount of wind power solved with robust optimization.	Fictious	A flexible model that ensures reliability and security.
[40]	Overview on formulations and optimization methods for the unit-based short-term hydro scheduling problem.	Norway	Fewer works summarize hydro scheduling works on single turbine models compared to aggregated models.
[41]	Acceleration of Benders decomposition for two-stage stochastic maintenance scheduling.	United Kingdom	The formulation approximates the three dimensional nonlinearity of hydropower production.
[42]	Stochastic two-stage NLP and MILP, inflow stochastic	Canada	Using multi-scenarios is more robust than using median scenarios.
[43]	The model is a combination of simulation and stochastic two-stage LP. Inflow and electricity price are stochastic.	Nordic Countries	The model is well suited to valuate the flexibility of a power system.

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